



UNIVERSITY OF PETROLEUM AND ENERGY  
STUDIES  
DEHRADUN

# **Applied DevOps**

## **Experiment 1**

**MTECH - School of Computer Science**  
**CYBER SECURITY AND FORENSICS**

Submitted To: Prof. KrishnaRaj

Name: Jigesh Sheoran

SAP ID: 590025428

**Objective — Design and implement a simple DevOps pipeline that automatically builds and tests an application whenever code is pushed to a version control repository.**

### **Problem Statement:**

In traditional software development, code changes are often tested manually, which can lead to delays, errors, and poor software quality. Modern software development adopts **DevOps practices** to automate testing and ensure rapid and reliable delivery.

### **Tools and Environment :**

- Git and GitHub (free account)
- GitHub Actions (CI tool)
- Python 3.12.0
- Pytest
- VS Code - Code Editor

### **Task to be Performed**

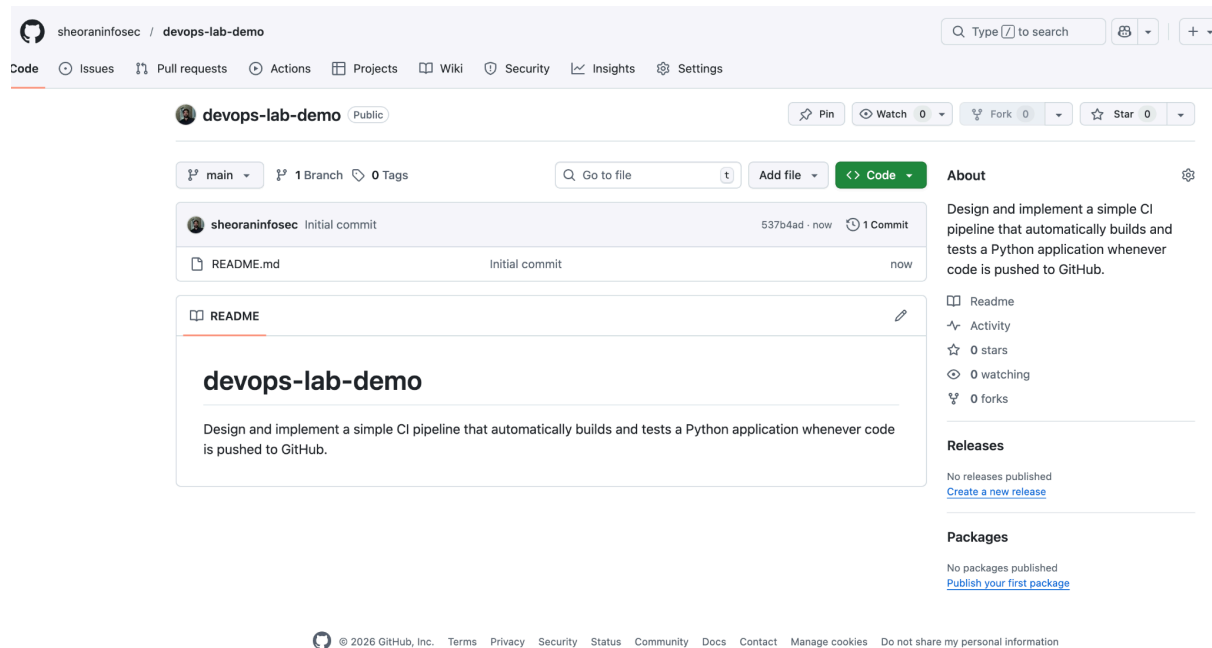
1. **Create a GitHub Repository** named devops-lab-demo.
2. **Develop a simple Python application** that performs an arithmetic operation (e.g., addition).
3. **Write an automated test case** to verify the correctness of the application.
  - **Configure a Continuous Integration (CI) pipeline** using **GitHub Actions** that: Automatically triggers on every code push
  - Installs required dependencies
  - Executes the test cases
  - Displays the test result (pass/fail)
  - 
  - **Demonstrate the DevOps workflow** by: Making a code change
  - Pushing it to GitHub
  -

Observing the automated execution of the pipeline

6. **Intentionally introduce an error** in the code and push it again to observe pipeline failure.

## Step-by-Step Procedure:

### 1. Create a new github repository named as “devops-lab-demo”



### 2. Clone this repository to your local workstation / environment.

```
devops-lab-demo — -zsh — 102x24

Last login: Wed Dec 17 23:41:53 on ttys000
sheoraninfosec@Jigeshs-MacBook-Air ~ % git clone https://github.com/sheoraninfosec/devops-lab-demo.git

Cloning into 'devops-lab-demo'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
sheoraninfosec@Jigeshs-MacBook-Air ~ % cd devops-lab-demo
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

### 3. Creating a simple python application that performs arithmetic operations like addition, subtraction, multiplication & division.

### 4. Here is the following **app.py**

The screenshot shows the VS Code editor interface. The Explorer sidebar on the left shows the project structure with 'DEVOPS-LAB-DEMO' containing 'app.py' and 'test\_app.py'. The main editor window displays 'app.py' with the following code:

```
1 # Date Created : 22nd January 2026
2 # Author: Jigesh Sheoran
3
4 def addition(a, b):
5     return a + b
6
7 def subtraction(a, b):
8     return a - b
9
10 def multiplication(a, b):
11     return a * b
12
13 def division(a, b):
14     return a / b
15
16 if __name__ == "__main__":
17     while True:
18         print("Calculator")
19         print("1. Add")
20         print("2. Subtract")
21         print("3. Multiply")
22         print("4. Divide")
23         print("5. Exit")
24
25         choice = int(input("Enter Operation: "))
26
27         if choice == 5:
28             print("Exiting calculator.")
29             break
30
31         a = float(input("Enter 1st number: "))
32         b = float(input("Enter 2nd number: "))
33
34         if choice == 1:
35             print(addition(a, b))
36         elif choice == 2:
37             print(subtraction(a, b))
38         elif choice == 3:
39             print(multiplication(a, b))
40         elif choice == 4:
41             print(division(a, b))
```

## 5. Now write test cases for this app.py

The screenshot shows the VS Code editor interface with 'test\_app.py' open. The Explorer sidebar shows 'test\_app.py' selected. The main editor window displays the following test cases:

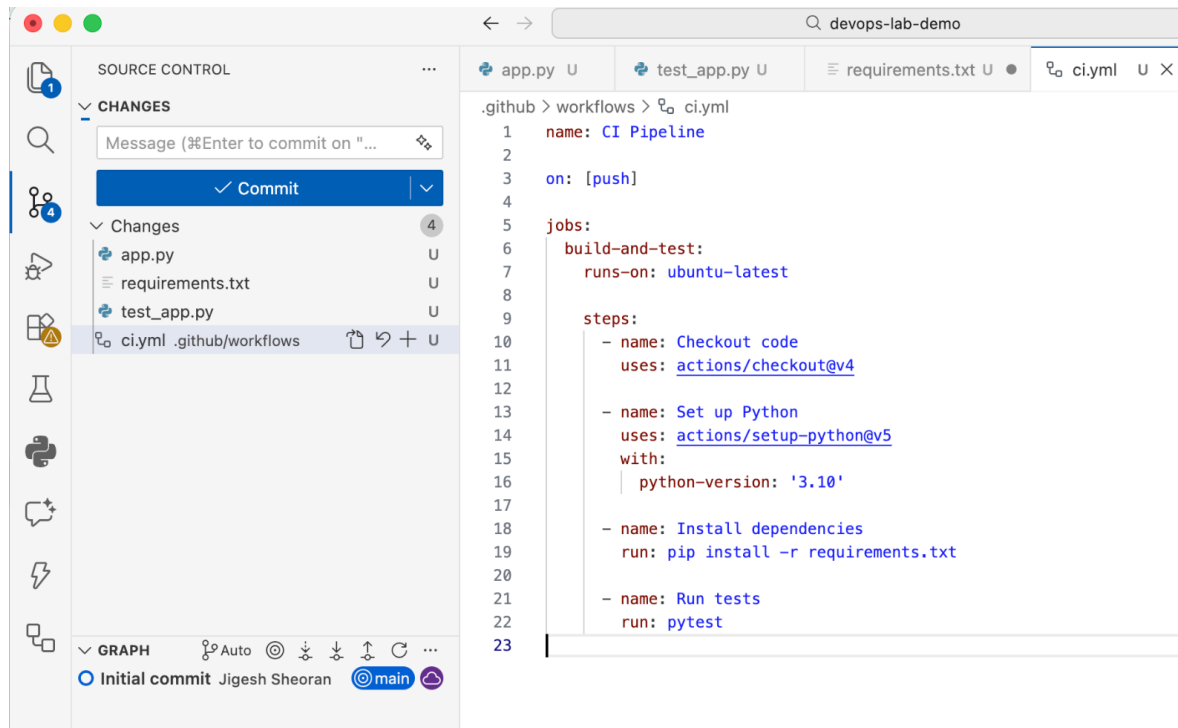
```
1 from app import addition, subtraction, multiplication, division
2
3 def test_addition():
4     assert addition(3, 3) == 6
5
6 def test_subtraction():
7     assert subtraction(7, 3) == 4
8
9 def test_multiplication():
10    assert multiplication(9, 3) == 27
11
12 def test_division():
13    assert division(30, 2) == 15
14
```

## 6. requirements.txt

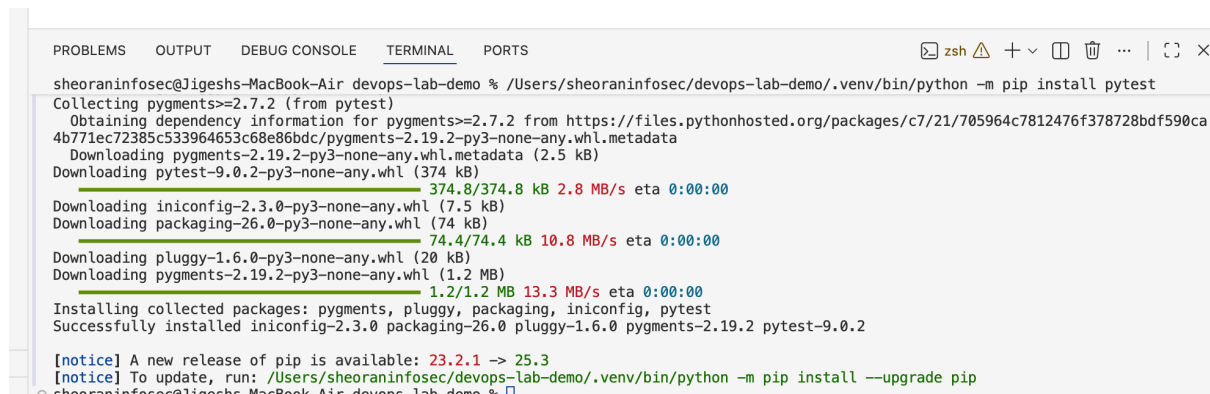
The screenshot shows the VS Code editor interface with 'requirements.txt' open. The Explorer sidebar shows 'requirements.txt' selected. The main editor window displays the following content:

```
1 ||| pytest
```

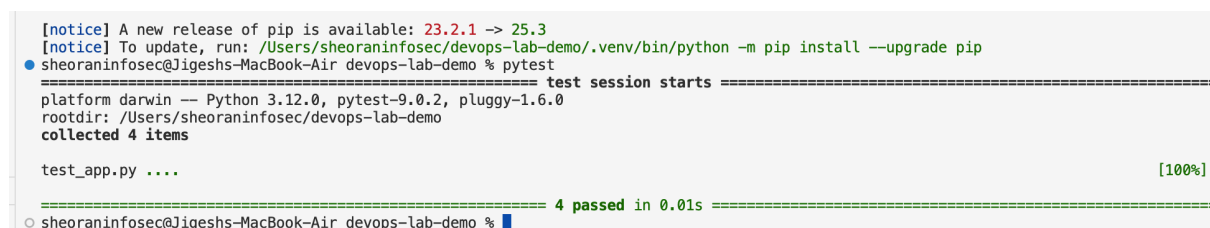
## 7. CI Workflow (.github/workflows/ci.yml)



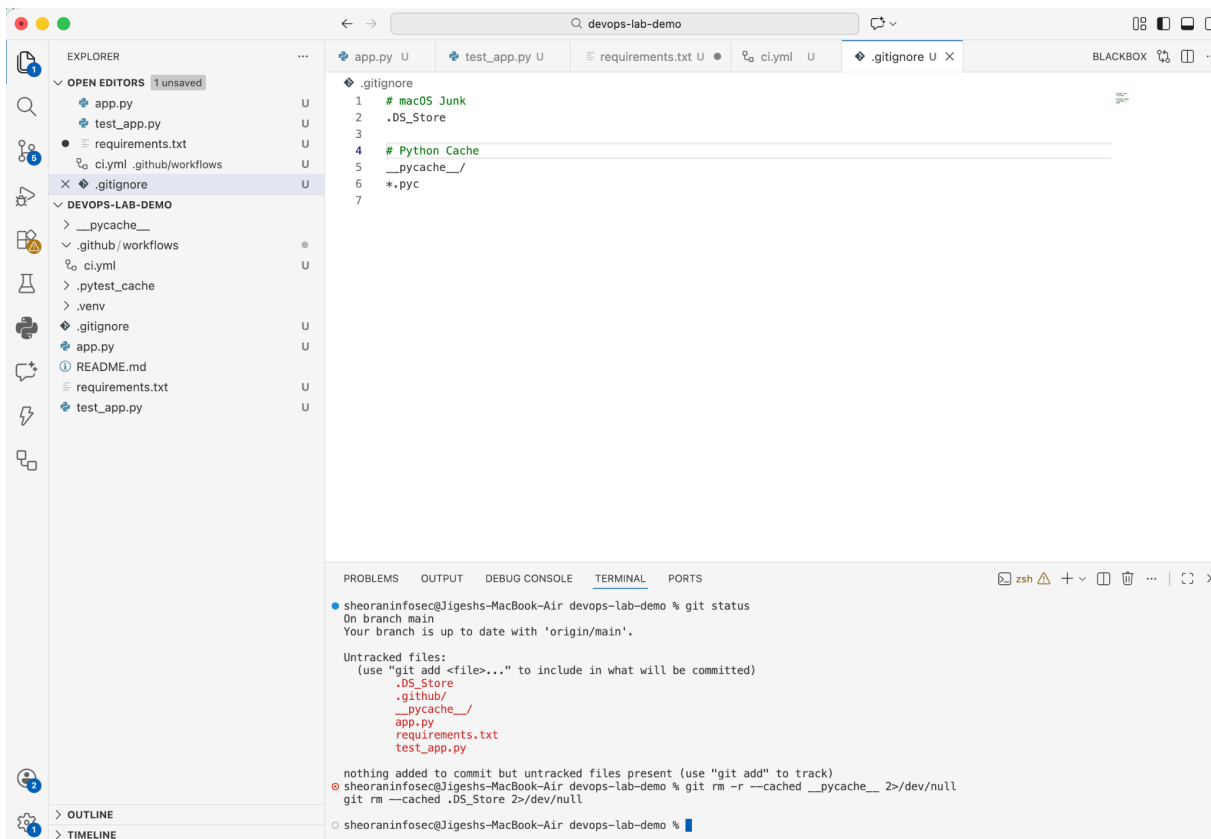
## 8. Installing pytest, now we will be able to run our test cases locally also.



## 9. Run “pytest” to run the test cases. In the above code we have written 4 test cases and as you can see that all the 4 test cases have been passed when we ran them locally.



10. I've made it a practice to add `.gitignore` in all of my codes because it helps to prevent unnecessary, system-generated and sensitive files from being uploaded to GitHub.



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows the file structure of a project named 'DEVOPS-LAB-DEMO'. The file list includes `__pycache__`, `.github/workflows`, `ci.yml`, `.pytest_cache`, `.venv`, `.gitignore`, `app.py`, `README.md`, `requirements.txt`, and `test_app.py`. The `.gitignore` file is selected, and its contents are displayed in the editor. The terminal window at the bottom shows the output of the `git status` command, indicating that the branch is up to date with 'origin/main' and listing untracked files: `.DS_Store`, `.github/`, `__pycache__/`, `app.py`, `requirements.txt`, and `test_app.py`. The terminal also shows the command `git rm --cached .DS_Store 2>/dev/null` being executed.

```
.gitignore
1 # macOS Junk
2 .DS_Store
3
4 # Python Cache
5 __pycache__/
6 *.pyc
7
```

```
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .DS_Store
        .github/
        __pycache__/
        app.py
        requirements.txt
        test_app.py

nothing added to commit but untracked files present (use "git add" to track)
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git rm --cached __pycache__ 2>/dev/null
git rm --cached .DS_Store 2>/dev/null
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

11. **git status** tells you what has changed, what is tracked, and what will be committed next.



The screenshot shows a terminal window with the output of the `git status` command. The output indicates that the branch is up to date with 'origin/main' and lists untracked files: `.DS_Store`, `.github/`, `__pycache__/`, `app.py`, `requirements.txt`, and `test_app.py`. The terminal also shows the command `git rm --cached .DS_Store 2>/dev/null` being executed.

```
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .DS_Store
        .github/
        __pycache__/
        app.py
        requirements.txt
        test_app.py

nothing added to commit but untracked files present (use "git add" to track)
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git rm --cached __pycache__ 2>/dev/null
git rm --cached .DS_Store 2>/dev/null
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

## 12. **git add** moves changes from your working directory to the staging area.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git add .
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   .github/workflows/ci.yml
        new file:   .gitignore
        new file:   app.py
        new file:   requirements.txt
        new file:   test_app.py

sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

## 13. **git commit -m** creates a snapshot of the staged files and records it in Git history.

```
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git commit -m "Add calculator app Github Actions"
[main fa61c14] Add calculator app Github Actions
5 files changed, 82 insertions(+)
create mode 100644 .github/workflows/ci.yml
create mode 100644 .gitignore
create mode 100644 app.py
create mode 100644 requirements.txt
create mode 100644 test_app.py
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

## 14. **git push origin main** uploads local commits from the main branch to the remote GitHub repository. But before that we need to set up secure authentication methods. (**SSH Key**) in this case.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git push origin main
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % ls ~/.ssh
id_rsa      id_rsa.pub  known_hosts  known_hosts.old
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % ssh-keygen -t ed25519 -C "jsheoran9755@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/Users/sheoraninfosec/.ssh/id_ed25519):
Enter passphrase for "/Users/sheoraninfosec/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /Users/sheoraninfosec/.ssh/id_ed25519
Your public key has been saved in /Users/sheoraninfosec/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:jffZa42Iqq3D2VfzT7Ww9HdTejXCCUbHt7om8Ld1nA jsheoran9755@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|      + .      |
|     o O .     |
|    * o .      |
|   o . o O*   |
|  S o . +.E   |
| o OO* =+    |
| . O OO*++++ |
| +.. O...++O |
| o+OO .++O   |
+-----[SHA256]-----+
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

## 15. To make connection of the local github to the Github server I have set up an SSH key, being a cybersecurity student that's my preferred authentication method.



**Jigesh Sheoran (sheoraninfosec)**  
Your personal account

[Go to your personal profile](#)

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

**SSH and GPG keys**

Organizations

Enterprises

Moderation

## SSH keys

New SSH key

There are no SSH keys associated with your account.

Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

## GPG keys

New GPG key

There are no GPG keys associated with your account.

Learn how to [generate a GPG key and add it to your account](#).

## Vigilant mode

### ☐ Flag unsigned commits as unverified

This will include any commit attributed to your account but not signed with your GPG or S/MIME key.

Note that this will include your existing unsigned commits.

[Learn about vigilant mode](#).

## Confirm access



Signed in as @sheoraninfosec



### Passkey

When you are ready, authenticate using the button below.

Use passkey

### Having problems?

- [Use your authenticator app](#)
- [Use your password](#)

**Tip:** You are entering [sudo mode](#). After you've performed a sudo-protected action, you'll only be asked to re-authenticate again after a few hours of inactivity.





## Jigesh Sheoran (sheoraninfosec)

Your personal account

[Go to your personal profile](#)

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

## SSH keys

[New SSH key](#)

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

### Authentication keys



Jigesh MacBook Air

SHA256: [redacted]

SSH

Added on Jan 22, 2026

Never used — Read/write

Delete

Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

16. You can see from the above screenshots how to generate a SSH key and use it to authenticate.

```
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % ssh -T git@github.com
Hi sheoraninfosec! You've successfully authenticated, but GitHub does not provide shell access.
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

17. **git remote -v** displays the remote repositories linked to your local Git repository along with their URLs.

```
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git remote -v
origin https://github.com/sheoraninfosec/devops-lab-demo.git (fetch)
origin https://github.com/sheoraninfosec/devops-lab-demo.git (push)
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git remote set-url origin git@github.com:sheoraninfosec/devops-lab-demo.git

sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git remote -v
origin git@github.com:sheoraninfosec/devops-lab-demo.git (fetch)
origin git@github.com:sheoraninfosec/devops-lab-demo.git (push)
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

18. Finally we can commit our code to the github server.

```
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo % git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 1.25 KiB | 1.25 MiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/devops-lab-demo.git
537b4ad..fa61c14 main -> main
sheoraninfosec@Jigeshs-MacBook-Air devops-lab-demo %
```

## 19.Executing the CI Pipeline.

← CI Pipeline

✖ Add calculator app Github Actions #1 Re-run jobs ...

Summary

All jobs

✖ build-and-test

Run details

Usage

Workflow file

Triggered via push 1 minute ago

sheoraninfosec pushed fa61c14 main

Status: Failure

Total duration: 10s

Artifacts: -

ci.yml

on: push

✖ build-and-test 6s

Annotations

1 error

✖ build-and-test

Process completed with exit code 127.

## 20.Fixed the CI Pipeline

All workflows

Showing runs from all workflows Filter workflow runs

2 workflow runs

|                                                         | Event | Status | Branch | Actor         |
|---------------------------------------------------------|-------|--------|--------|---------------|
| ✓ Update requirements.txt                               |       |        | main   |               |
| CI Pipeline #2: Commit e5b6ab8 pushed by sheoraninfosec |       |        |        |               |
|                                                         |       |        |        | 1 minute ago  |
|                                                         |       |        |        | 15s           |
|                                                         |       |        |        | ...           |
| ✖ Add calculator app Github Actions                     |       |        | main   |               |
| CI Pipeline #1: Commit fa61c14 pushed by sheoraninfosec |       |        |        |               |
|                                                         |       |        |        | 7 minutes ago |
|                                                         |       |        |        | 10s           |
|                                                         |       |        |        | ...           |

## 21.Now intentionally committing the faulty code which will fail the test cases and will ultimately result in failing the CI Pipeline.

All workflows

Showing runs from all workflows Filter workflow runs

2 workflow runs

|                                                         | Event | Status | Branch | Actor          |
|---------------------------------------------------------|-------|--------|--------|----------------|
| ✖ fix subtraction logic                                 |       |        | main   |                |
| CI Pipeline #5: Commit 18e6ce7 pushed by sheoraninfosec |       |        |        |                |
|                                                         |       |        |        | 48 minutes ago |
|                                                         |       |        |        | 9s             |
|                                                         |       |        |        | ...            |
| ✖ bug in subtraction                                    |       |        | main   |                |
| CI Pipeline #4: Commit 08812e5 pushed by sheoraninfosec |       |        |        |                |
|                                                         |       |        |        | 53 minutes ago |
|                                                         |       |        |        | 12s            |
|                                                         |       |        |        | ...            |

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